

Indian Institute of Engineering Science and Technology, Shibpur
B.Tech (3rd Semester, CST) Mid-Semester Examination, September 2024
Digital Logic (CS 2102)

F.M. 30

Answer any 3 questions

Time: 2 hours

1. (a) Determine the possible unknown base b of the following relation: $(\sqrt{22})_b = (6)_{10}$.
- (b) Represent the following decimal numbers in BCD and perform the BCD addition. The two numbers are 28 and 39.
- (c) Assume that in a combinatorial circuit with output F , the minterms can be given as below, where m denotes the minterms and d denotes the don't care conditionals.

$$f(a, b, c, d) = \sum m(0, 2, 5, 7, 9, 11) + \sum d(3, 8, 10, 12, 14)$$

Derive all the prime implicants and essential prime implicants using K-map. [2 + 3 + 5]

2. Design a combinational circuit with three inputs x, y, z , and has three outputs A, B , and C . When the binary input is 0, 1, 2 or 3, the binary output is one greater than the input. When the binary input is 4, 5, 6, or 7, the binary output is one less than the input. [10]
3. (a) Consider the combinational circuit given below. Derive the minimal sum-of-product expression for the output F .

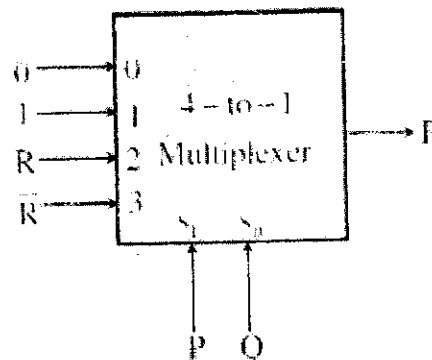


Figure 1: A combinatorial implemented using MUX.

- (b) Using four XOR gates and minimum number of full-adders, construct a 4-bit parallel adder/subtractor circuit. Use an input select variable S so that when $S = 0$, the circuit adds and when $S = 1$, the circuit subtracts. [6 + 4]
4. (a) Write the characteristic table and characteristic equation of $J - K$ flip-flop. Also implement the $J - K$ flip-flop using two-input NAND gates only.
- (b) Is there any problem in $J - K$ flip-flop? If yes, what are the methods to solve that problem. [6 + 4]